

Variants of DnaB in *Bacillus Subtilis*

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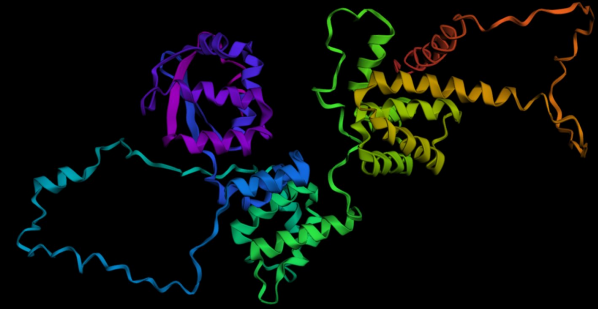
Introduction to DnaB

- Primarily present in the cytoplasm and cell membrane of *Bacillus subtilis* ¹
- Domains: DDBH1, DDBH2-1, DDBH2-2 ¹
- DnaC is the replicative helicase: DnaB, interacts with DnaD to 'load' DnaC onto *oriC* ^{1,2}
- Also facilitates the unwinding of DNA ¹
- Restarts stalled replications ³

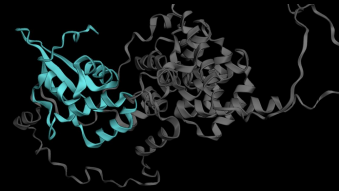
1: <https://www.uniprot.org/uniprotkb/P07908/>

2: <https://doi.org/10.1111/j.1365-2958.2004.04091.x>

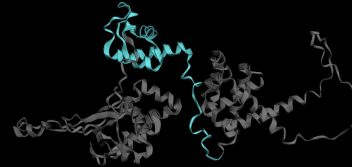
3: <https://doi.org/10.1046/j.1365-2958.2001.02631.x>



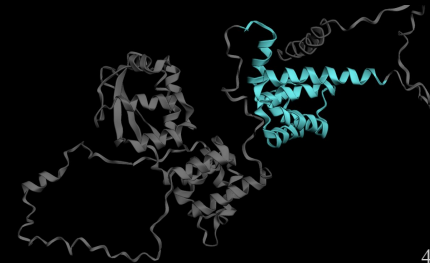
1. ColabFold Structure of DnaB



2. DDBH1



3. DDBH2-1



4. DDBH2-2

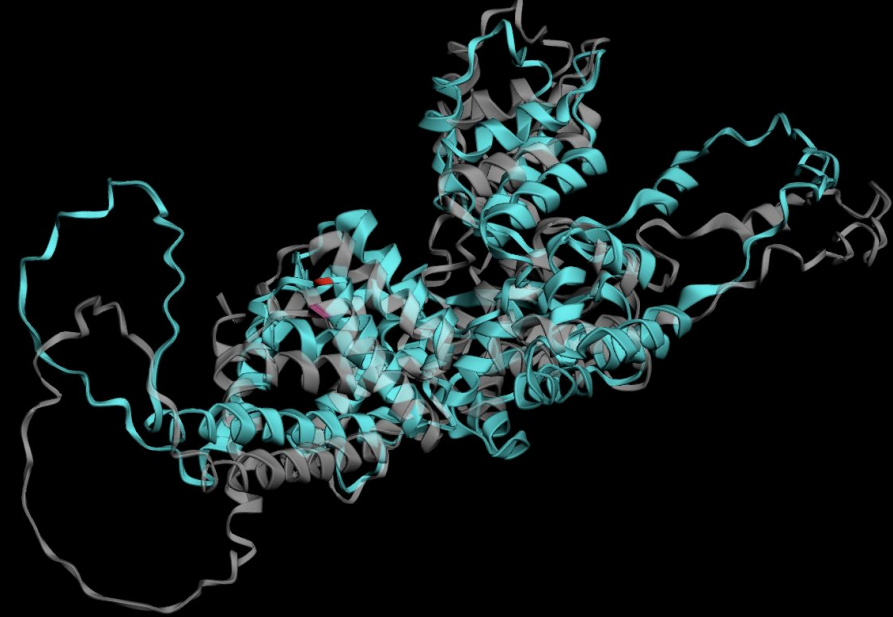
DnaBS371P

- Point Mutation at DDBH2-2: 371S → 371P
- Conformational Change: Global fold stays the same, but local deviations are observed
- Increased ssDNA affinity ¹
- Increased unwinding activity ^{1,2}
- Aspecific DnaC loading ³
- Aberrant Replication initiation and mediation ²

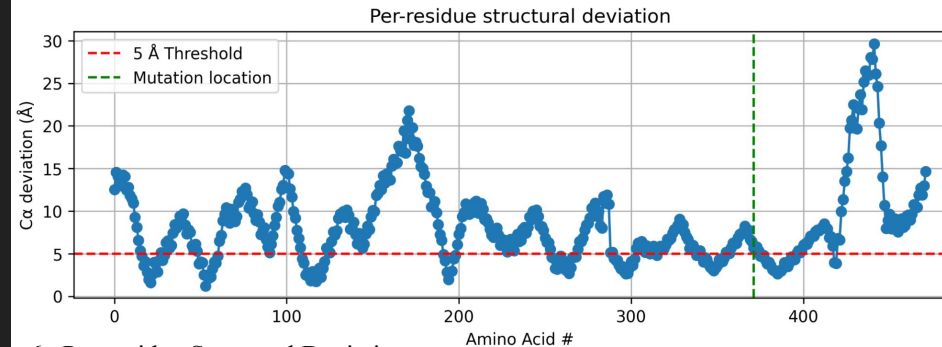
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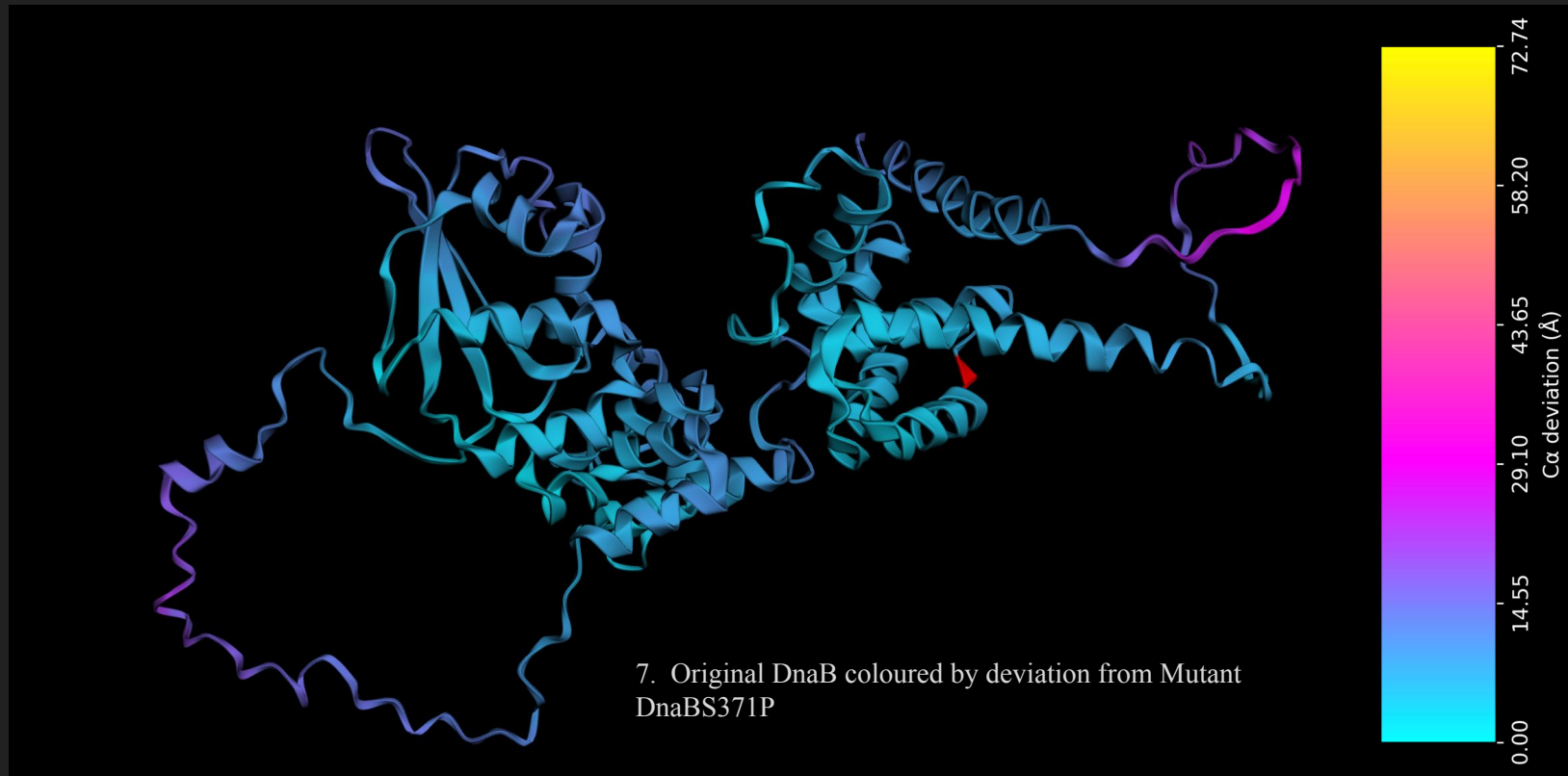


5. DnaB Original structure superimposed on Mutant



6. Per-residue Structural Deviation

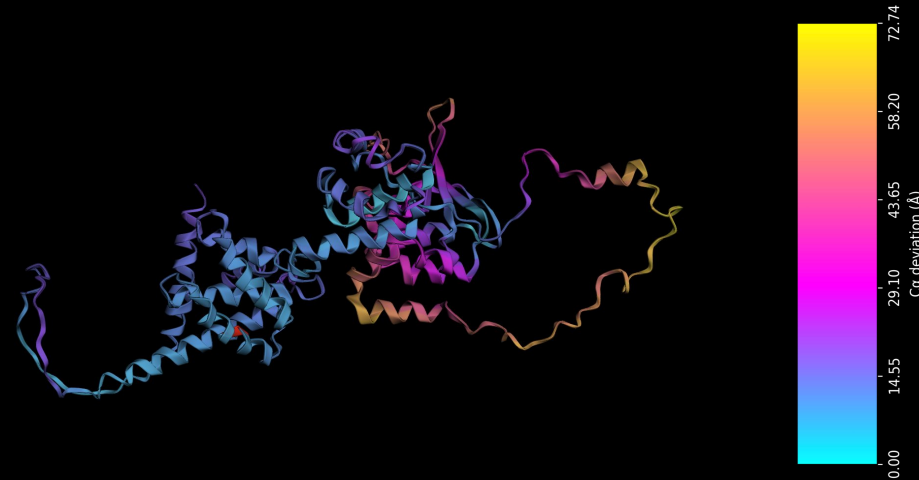
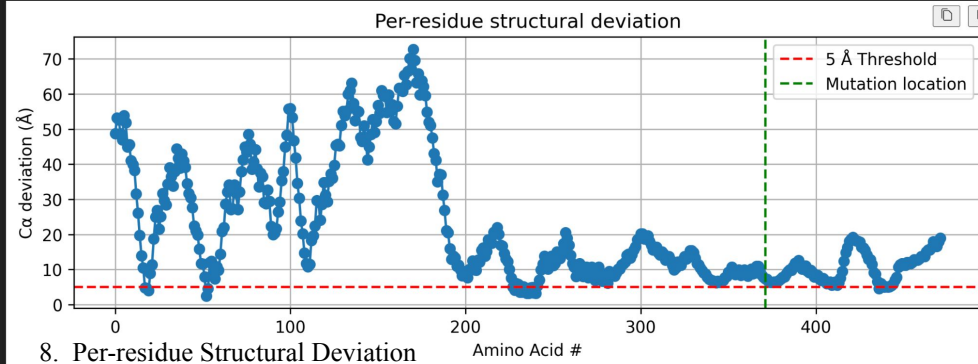
DnaB371P: Deviation from Wild-type



Another Mutation:

- Point Mutation at DDBH2-2: 379Y → 379A
- Conformational Change: Similar to DnaBS371P, but greater deviations observed
- Replication Initiation inhibited at 45°C₄

4: <https://pmc.ncbi.nlm.nih.gov/articles/PMC304273/>



9. DnaB Original structure coloured by deviations from Mutant